little progress has been made in this country regarding adoption of steel staybolts instead of iron.

At the present time steel staybolts are used on but 2 percent of all the locomotives investigated as mentioned above, the total number investigated being 88 percent, and information received thereon should be representative of all the locomotives in the country, this report covering 57,900 locomotives out of a total of 65,900.

The majority of installations of steel staybolts are still reported as being on test. In a few cases where opinions are expressed they are favorable, and one southern road of considerable size has adopted the steel staybolts as standard, the number of locomotives on this road being 80 percent of the total of all those using steel staybolts.

To make this clearer, steel staybolts have been adopted as standard on one road having over 1000 locomotives, and these are 80 percent of the 2 percent of all locomotives in the United States equipped with steel staybolts.

It would appear at the present time that there are not enough conclusive results available to express a definite opinion as regards the benefits derived from the use of steel staybolts as compared with iron, in the United States.

The accompanying table shows typical specifications for steel staybolts in five different cases.

In England steel staybolts are in general use, also staybolts of iron and copper, and the results obtained with steel staybolts are reported as being satisfactory and very favorable.

In Canada steel staybolts have been used for six or seven years on one of the large roads with excellent results, and on another large trunk line both steel and iron bolts are used and it is expected that steel will be adopted as standard.

In Mexico inquiries covering 411 locomotives elicited reports that steel staybolts are not used.

Regarding the benefits derived from the use of rigid hollow bolts instead of solid: These are undoubtedly beneficial, being much easier for inspection purposes as indicating broken bolts no matter where the fracture is located and save time, in that it is not necessary to remove any grate bars or brick work to make the ordinary hammer inspection.

A great many roads are going to the use of hollow flexible staybolts instead of solid flexibles, as with these any failed bolts can readily be detected as soon as the fracture occurs.

With this type of staybolt it is also possible to make use of the electrical inspection method thereby eliminating the necessity of removing caps every two years.

This report was prepared by a committee consisting of W. R. Hedeman, chairman; R. A. Pearson, and Lewis Nicholas.

COMPARISON OF SPECIFICATIONS—STEEL STAYBOLTS

Specification	Carbon Percent	Percent Manganese	Silicon Percent	Sulphur Percent	Percent Phosphorous
A	.15 Max.	—	—	—	—
B	.10 to .12	.35 to .50	.15 to .20	.05 Max.	.05 Max.
C	.12 to .18	—	—	.06 Max.	.06 Max.
D	—	—	—	—	—
E	.10% Max.	—	—	.05% Max.	.05% Max.
Nickel Percent	Chromium Percent	Tensile Strength	Elastic Limit	Elongation in 8 in.	Reduction in Area
1.50 to 2.00	.15 to .30	69360	48930	32.75%	63.70%
—	—	55000 to 65000	65% of Tensile	30 to 35%	55 to 65%
—	—	64000 to 74000	—	28 to 23 over 3"	—
—	—	52000 to 60000	—	27%	—
—	—	50000 Min.	—	30% Min.	65% Min.

Discussion

I. J. POOL, (Baltimore & Ohio Railroad): My experience with steel staybolts is confined to one full installation of flexible steel staybolts applied to a Mallet compound engine with a working pressure of 220 pounds per square inch. The firebox end of these staybolts had a tell-tale hole and instead of driving the bolts in the usual way, they were expanded tight in their holes after which the edges of bolt heads were beaded down to the sheet with a special made beading tool.

I had an opportunity to follow this installation of staybolts for about two years during which time the locomotive ran approximately 40,000 miles. During this period our inspections indicated that we experienced considerably less staybolt leakage in fire line of side sheets than on other locomotives operating under similar conditions. I attribute this improvement to the staybolts being expanded instead of driven.

Some time later we attempted to apply some iron staybolts in the same manner as the steel bolts were applied. In doing this we found the iron would split before bolt would expand tight in hole and cause a bad condition. After the engine, I referred to, had been in service for approximately one year, we began experiencing considerable trouble with the staybolts breaking. Examination of staybolts adjacent to those found broken showed no evidence of fracture such as is frequently found in iron staybolts. This condition left me with the impression that steel staybolts when found broken fail without giving warning and do not develop the same kind of progressive fractures as do iron staybolts. In connection with the use of hollow staybolts—this type bolt is becoming more popular for the reason that it simplifies inspection. The fact that a fracture can be located anywhere regardless of location is an item well worth consideration.

R. A. PEARSON, (Canadian Pacific Railway): In our experience steel staybolts have less breakage than iron. Hollow flexible bolts are gradually coming into wider use on our railroad and give excellent satisfaction. We have not had steel staybolts in service long enough yet to determine exactly how they will work out.

M. V. MILTON, (Canadian National Railway): Five years ago we built 100 engines equipped with steel staybolts, both solid and hollow flexible type. On one mountain-type locomotive we have had a mileage of 300,000 with only one bolt breaking. All later power is being equipped with steel staybolts. The steel has a small nickel content.

W. H. LAUGHRIDGE, (Hocking Valley Railroad): We have not had enough experience as yet with steel staybolts to make any definite statement in this connection. We have, however, experienced difficulty in threading the steel bolts satisfactorily and finally solved the problem by grinding the dies carefully and threading the bolts while maintaining sufficient and proper lubrication. Seven years ago we tried out an old ogee firebox with steel bolts and ran it for two years with complete satisfaction. Recently we fitted a Mallet with a complete installation of steel bolts. We expect good results from the use of steel.

E. P. FAIRCHILD, (Atlantic Coast Line): We have